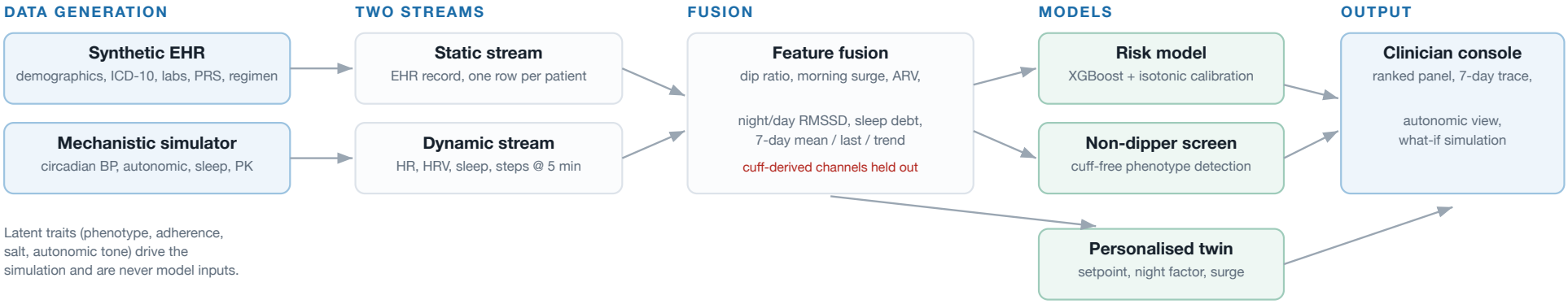
CARDIOTWIN · SYSTEM ARCHITECTURE

From synthetic cohort to a clinician decision



Latent traits (phenotype, adherence, salt, autonomic tone) drive the simulation and are never model inputs.

COUNTERFACTUAL VALIDATION LOOP

The twin predicts the change from an intervention. The ground-truth simulator is re-run for the same virtual patient, with the same random seed and the same override. The two are compared as a paired experiment, so the what-if engine is measured rather than asserted.

twin.predict_change(i) → predicted ΔSBP vs simulate_patient(overrides=i) → actual ΔSBP reported as MAE and sign agreement

DEPLOYMENT PATH

one calibration week with a cuff → twin personalised → continuous cuff-free monitoring from the wearable → cuff or ABPM requested only when the twin flags risk